

- 3 The tortoise beetle, *Chelymorphism alternans*, is an insect found in Panama that has several different colour patterns. Fig. 3.1 shows a tortoise beetle.

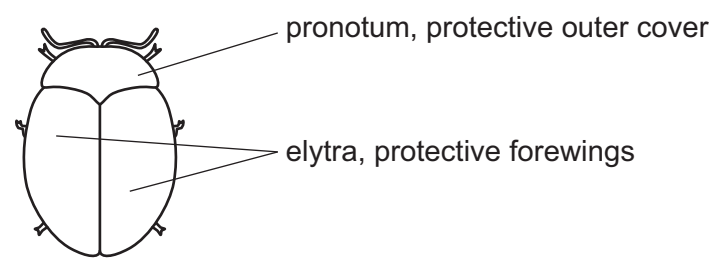
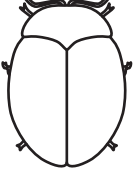


Fig. 3.1

Researchers have identified a gene, **L**, that controls colour pattern in the pronotum and elytra. Gene **L** has four different alleles: **L^V**, **L^T**, **L^R** and **L^r**.

Table 3.1 shows five different colour pattern phenotypes of tortoise beetles and their genotypes.

Table 3.1

phenotype	genotypes
 black red rufipennis	L^R L^R L^R L^V L^R L^r
 black and red striped darien f. militaris-a (dfm-a)	L^T L^T L^T L^V L^T L^r
 black and red striped darien f. militaris-b (dfm-b)	L^T L^R
 red veraguensis	L^V L^V L^V L^r
 metallic and red striped metallic	L^r L^r



- (a) Explain why the inheritance of colour pattern in tortoise beetles can be described as involving multiple alleles.

.....
.....
..... [1]

- (b) A tortoise beetle with **dfm-b** phenotype was crossed with another tortoise beetle with **dfm-b** phenotype.

Construct a genetic diagram to show the results of this cross, including the ratio of offspring phenotypes.

parental phenotypes: **dfm-b** × **dfm-b**

parental genotypes:

gametes:

offspring genotypes:

offspring phenotypes:

ratio of offspring phenotypes:

[4]

- (c) Colour pattern phenotype involves alleles that show codominance. There is also an order of dominance of alleles (dominance hierarchy).

Use the information in Table 3.1 to:

- identify the codominant alleles
- list the dominance hierarchy with alleles from the most dominant to the least dominant.

codominant alleles

dominance hierarchy

[2]



(d) Researchers carried out two crosses.

Cross 1: female veraguensis tortoise beetles were crossed with male metallic tortoise beetles.

The results are shown in Table 3.2.

Table 3.2

number of observed offspring phenotypes				ratio of observed colour pattern phenotypes
male veraguensis	female veraguensis	male metallic	female metallic	
139	153	136	140	1.06 : 1

Cross 2: female veraguensis tortoise beetles were crossed with male veraguensis tortoise beetles.

The results are shown in Table 3.3.

Table 3.3

number of observed offspring phenotypes				ratio of observed colour pattern phenotypes
male veraguensis	female veraguensis	male metallic	female metallic	
693	592	237	213	2.9 : 1

- (i) Using Table 3.1, deduce the genotypes of **each** of the parental beetles used in cross 1 and cross 2.

cross 1

cross 2

[2]

- (ii) An assumption was made that female tortoise beetles have XX chromosomes and males have XY chromosomes. Gene **L** is **not** located on the X chromosome. It was concluded that colour pattern phenotype followed autosomal inheritance.

Explain how the evidence in Table 3.3 supports this conclusion.

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.....

..... [1]

- (iii) Name a statistical test that can be used to determine whether the ratio of observed phenotypes is significantly different from the expected ratio.

..... [1]

[Total: 11]